Chapter 2

INTRODUCTION TO ALGEBRA

A8IABP

Learning Objectives

- To describe, extend, create numeric and geometric patterns.
- To make predictions related to the patterns and investigate repeating patterns.
- To understand the role of 'variables' in patterns.
- To use variables in simple algebraic expressions and equations to describe relationships.

2.1 Introduction

Are you ready for a number game? Follow the steps carefully:

Step 1	Step 2	Step 3	Step 4	Step 5
Think of any number	Multiply it by 2	Add 20	Divide by 2	Subtract the original number you had thought in step 1

Is your answer 10? Is it the same for all in the class? Verify it with your friend who might have started with a number other than your number. Surprised? What if you started with a fraction, say $\frac{1}{2}$ or $\frac{3}{4}$ or $\frac{4}{5}$? In this game, regardless of the number you started with, the answer will be 10.

Let us verify the game for two more numbers, say 4 and 9.

- If the initial number is 4,

Step 1	Step 2	Step 3	Step 4	Step 5
4	4 × 2 = 8	8 + 20 = 28	28 ÷ 2 = 14	14 – 4 = 10

- If the initial number is 9,

Step 1	Step 2	Step 3	Step 4	Step 5
9	9 × 2 = 18	18 + 20 = 38	38 ÷ 2 = 19	19 – 9 = 10

6th_Chapter 2_032-045.indd 32 1/27/2020 1:42:55 PM